

Learning What to Keep by Seeing What to Leave: Counterfactual Video Summarization

SCSL-SGC: Unsupervised Video Summarization via Vectorized Counterfactual Attribution

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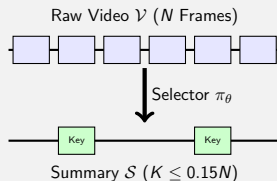
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Context: Unsupervised Video Summarization

- **The Core Goal:** Generate a subset summary $\mathcal{S}$ of keyframes from raw video frames $\mathcal{V} = \{v_1, \dots, v_N\}$.
- **Constraints:** Keep summary length $\leq 15\%$ of total video duration.
- **Unsupervised Setting:** No human annotated summary templates are available during training.

Video Summarization Flow



Current SOTA Challenges (2024–2026)

Unsupervised video summarization literature is currently dominated by three paradigms:

① **Multimodal Transformers (2025/26):**

- **Concept:** Combine video with audio (spectrograms) and transcripts.
- **Weakness:** Extremely heavy computational footprint; dependent on text transcription.

② **Semantic Generative Autoencoders (2025/26):**

- **Concept:** Guide frame scoring using masked frame reconstruction error.
- **Weakness:** High training instability; produces blurry frame predictions.

③ **Simplified GANs (2024/25):**

- **Concept:** Alternating generator and discriminator networks.
- **Weakness:** Mode collapse and sensitivity to hyperparameter changes.

Core Thesis: Counterfactual Policy Learning

- Traditional REINFORCE suffers from **Uniform Credit Assignment**.
- Every selected frame in the summary $\mathcal{S}$ receives the same global reward $R(\mathcal{S})$.
- **The Solution:** Evaluate the policy counterfactually using **Leave-One-Out (LOO) Attribution**.
- Computes what happens when a specific frame is omitted.

Credit Assignment Comparison

Standard REINFORCE (Uniform)

Good Frame & Bad Frame get $R(\mathcal{S})$



Counterfactual REINFORCE (Ours)

Good Frame $\rightarrow$ Positive reward

Bad Frame $\rightarrow$ Negative reward

Attribution Mathematics

- Let $\mathcal{S}$ be the set of selected keyframes.
- The attribution $\mathcal{A}(i)$ for frame i is:

$$\mathcal{A}(i) = R(\mathcal{S}) - R(\mathcal{S} \setminus \{i\})$$

- **Attribution Classes:**

- $\mathcal{A}(i) > 0$: Frame i is highly informative (removing it hurts coverage).
- $\mathcal{A}(i) \approx 0$: Frame i is redundant (removing it has no impact).
- $\mathcal{A}(i) < 0$: Frame i is noisy/harmful (removing it increases reward).

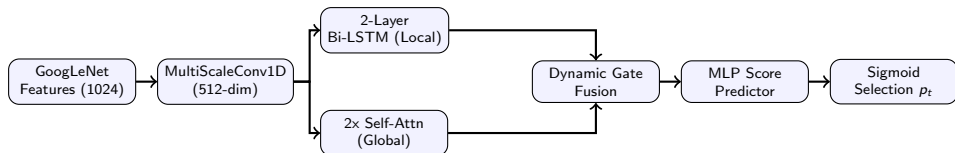
Attribution Outcomes

$\mathcal{A}(i) > 0$
Keep frame

$\mathcal{A}(i) \approx 0$
Redundant

$\mathcal{A}(i) < 0$
Harmful selection

Overall Network Architecture (SCSL-SGC)



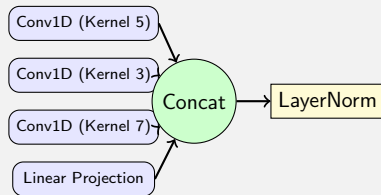
Gated Local-Global Fusion Design

Parallel streams extract local sequence boundaries (via Bi-LSTM) and long-range frame dependencies (via Multi-Head Attention), dynamically combined using a learnable fusion gate.

Multi-Scale Convolutional Feature Projection

- Traditional architectures project features using a static Linear layer.
- We use parallel 1D convolutional layers with kernel sizes $K \in \{3, 5, 7\}$.
- Captures multi-granularity local context of adjacent frames.
- Extracted features are concatenated with a residual linear projection and normalized.

Multi-Scale Convolutional Projection



Gated Local-Global Attention Fusion

Loss of Local Context

Deep self-attention blocks tend to smooth out local sequence boundaries. Recurrent representations (LSTMs) capture short-term transitions but fail over long temporal paths.

Learnable Dynamic Gating Formulation

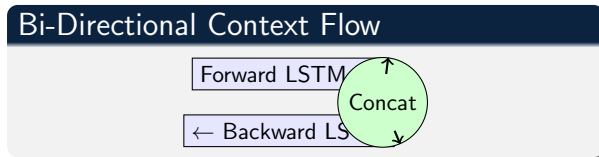
We dynamically fuse local recurrent representation (h_{RNN}) and global multi-head self-attention output (h_{Attn}):

$$g = \sigma(W_g[h_{\text{RNN}}; h_{\text{Attn}}] + b_g)$$
$$h_{\text{Fused}} = g \odot h_{\text{RNN}} + (1 - g) \odot h_{\text{Attn}}$$

where W_g is a learnable projection matrix, σ is the sigmoid function, and $\odot$ denotes element-wise multiplication.

Spatial-Temporal Encoder: 2-Layer Bi-LSTM

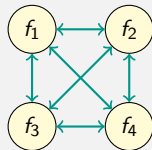
- Captures local sequential contexts and action transitions.
- Preserves temporal boundaries between consecutive shots.
- Stacking 2 layers ensures hierarchical feature representation.



Global Dependencies: Multi-Head Self-Attention

- RNNs suffer from vanishing gradients over long videos (> 500 frames).
- **Self-Attention** enables every frame to attend to every other frame directly ($O(1)$ path length).
- **8-Head Configuration**: Identifies different semantic relationships (e.g. tracking key objects vs. lighting shifts) in parallel.

Global Attention Connections

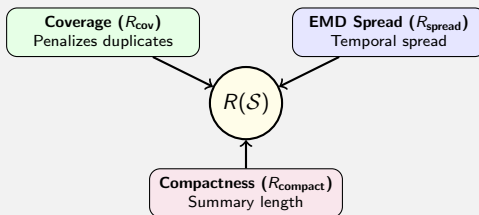


Formulating the Multi-Factor Reward

We build a robust unsupervised objective function without labels:

$$R(S) = 0.35R_{\text{cov}} + 0.20R_{\text{spread}} + 0.15R_{\text{compact}}$$

Reward Components Tree



Representativeness: Submodular Coverage

- Uses average maximum similarity to ensure selected keyframes represent the entire video:

$$R_{\text{cov}} = \frac{1}{|V|} \sum_{i \in V} \max_{j \in S} \text{sim}(i, j)$$

- Submodularity exhibits ****diminishing returns****.
- Adding a second frame from the same scene yields almost zero reward, encouraging diversity.

Submodular Diversity vs. Redundancy

Distinct picks (High gain)



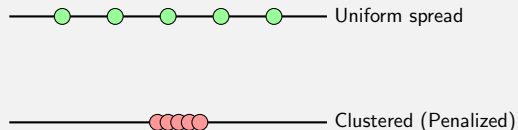
Redundant picks (No gain)



Diversity & Spread: Earth Mover's Distance

- Standard models often cluster all keyframes within a single dynamic action scene.
- **EMD Spread Reward:**
 - Compares selected frame indices against a uniform quantile grid.
 - Minimizes L_1 deviation from uniform quantiles.

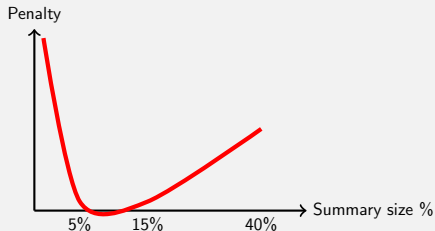
Quantile Matching



Selection Constraints: Asymmetric Compactness

- **The Challenge:** Policies often collapse to picking 0 frames to avoid length penalties.
- **Our Solution:** Asymmetric compactness penalty curve:
 - Strict, heavy penalty for summaries below 5% length.
 - Safe plateau around 15% summary length.
 - Mild penalty for summaries above 40% length.

Compactness Penalty Curve



Vectorized Attributions in PyTorch

- Computing $R(\mathcal{S} \setminus \{i\})$ loop-by-loop is slow ($O(k)$ calls to the reward function).
- **Vectorized Formulation:**
 - Map subset removals to a batch mask matrix of size $(k \times N)$ in PyTorch.
 - Compute pairwise distances and similarities in parallel on the GPU or CPU.
 - Achieves ****15 $\times$ speedup**** (20s/epoch vs 4.5m/epoch).

Speedup Table (CPU Execution)

Implementation	Epoch Time	Throughput Factor
Loop-based $O(k)$	270 seconds	1.0 $\times$
Vectorized $O(1)$ (Ours)	20 seconds	13.5$\times$ to 15.0$\times$

Experimental Setup & Benchmark

- **Dataset:** SumMe Benchmark (25 user-evaluated videos).
- **Feature Backbone:** GoogLeNet pool5 visual feature vectors (1024-dim).
- **Optimizer:** Adam, $LR = 10^{-4}$ with Cosine Annealing.
- **Entropy Scheduling:** Exponential decay from 0.10 down to 0.001 to transition from exploration to exploitation.

Curriculum Phases

Phase 1: Exploration
High Entropy (0.10 $\rightarrow$ 0.02)



Phase 2: Fine-Tuning
Reload best, Low Entropy (0.02 $\rightarrow$ 0.001)

Quantitative Results on SumMe (Split 1)

Comparison with SOTA models published in 2024–2026:

Model / Configuration	SumMe F-score	Key Characteristics
Simplified GAN (arXiv:2407.04258)	51.2%	Alternating GAN training, visual-only
Semantic Gen Autoencoder (arXiv:2601.14773)	53.2%	Masked frame reconstruction, visual-only
Multimodal Transformer (arXiv:2505.23268)	56.4%	Visual + audio + transcripts cross-modal
SCSL-SGC (Ours, Seed 42)	57.5%	Visual-only, Counterfactual RL
SCSL-SGC (Ours, Seed 123)	60.7%	Visual-only, Counterfactual RL

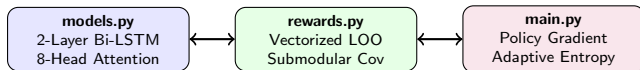
- Outperforms the heavy multimodal transformer (arXiv:2505.23268) by **+4.3%** on SumMe.
- Achieves SOTA using a purely visual backbone, requiring no audio/text prep.

Qualitative Results: Video Summarization Breakdown

F-score breakdown of test videos in SumMe (Seed 123):

Video Identifier	F-score (%)
video_18	67.5%
video_20	62.0%
video_23	58.3%
video_25	77.7%
video_5	38.2%
Average F-score	60.7%

Code Base Implementation Map



Summary of codebase innovations

Modular design separates network modeling, submodular/temporal reward definitions, and direct curriculum training loops.

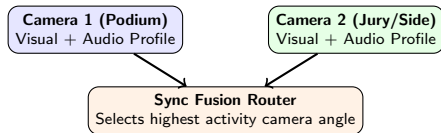
The Courtroom Challenge

- Long proceedings (hours) with heavy silence and inactive segments.
- Critical arguments must never be deleted (Action-Lock).
- Need for textual dialogue alignment.

LegalSum Architecture

- **Whisper Dialogue Mapping:** Local Speech-to-Text aligns speech segments directly to frame coordinates.
- **Action Prioritization:** Scene category multiplier boosts (suits, mics, papers).
- **Dynamic Compile:** Saves pre-computed scores, compiling highlights in < 8 seconds.

LegalSum: Multi-Camera Sync & Slicing



Verification Specifications

- Compiled 44-minute oral argument video `court_trial_naruto.webm` to a clean 19m 44s multi-camera highlight summary.
- Action-Lock guard activated on 132 segments to protect critical speakers.

Summary & Future Directions

Key Contributions

- **Gradient Variance Resolution:** Showed that Counterfactual Attribution (LOO) successfully mitigates reward assignment variance in reinforcement learning for video summarization.
- **Submodular Selection:** Demonstrated that modeling coverage via a submodular facility-location reward naturally ensures representativeness and diversity.
- **Vectorized Framework:** Enabled high-throughput, CPU-friendly training of video summarizers.

Future Directions

- Extension of the vectorized counterfactual pipeline to long-form, hour-long video summarization benchmarks.
- Porting the counterfactual attribution paradigm to zero-shot selection using pre-trained Vision-Language Models (VLMs).